

Career Roadmap: DevOps Engineer

Generated by AI Career Coach | 6 Month Plan

Week 1: Advanced Linux Administration

Deepen understanding of Linux internals, file systems, and kernel parameters to move beyond basic server management.

- Configure advanced systemd services with resource limits (cgroups). (3h) - Intermediate
- Audit system security using Lynis and fix top 3 vulnerabilities. (2h) - Intermediate

Week 2: Advanced Shell Scripting

Enhance Bash skills to automate complex administrative tasks and system checks.

- Write a Bash script to parse Nginx logs and output top IP addresses. (2h) - Intermediate
- Create an automated database backup script with cron and alert on failure. (3h) - Beginner

Week 3: Python for Systems Automation

Leverage existing Python skills to interact with OS-level APIs and cloud SDKs.

- Write a Python script using boto3 to find and stop unattached EC2 instances. (3h) - Intermediate
- Create a Python CLI tool to monitor disk usage and send a Slack webhook alert. (3h) - Intermediate

Week 4: Networking Deep Dive

Solidify OSI model, TCP/IP, DNS, and advanced routing concepts essential for cloud networks.

- Use tcpdump to capture and analyze HTTP traffic on a local VM. (2h) - Intermediate
- Configure a custom DNS server using bind9 on a local network. (3h) - Advanced

Week 5: AWS Networking & VPC Mastery

Expand basic AWS knowledge into complex network architectures.

- Provision two separate VPCs and establish a VPC peering connection between them using Terraform. (4h) - Intermediate
- Configure VPC flow logs and query them using Amazon Athena. (3h) - Advanced

Week 6: AWS Security & IAM

Transition from basic IAM to advanced least-privilege architectures and identity federation.

- Set up cross-account IAM roles for deployment automation. (3h) - Advanced
- Migrate hardcoded database credentials in your Python app to AWS Secrets Manager. (2h) - Intermediate

Week 7: Docker Mastery

Go beyond basic Dockerfiles to multi-stage builds, security scanning, and optimization.

- Refactor an existing Java/Spring Boot Dockerfile to use multi-stage builds, reducing image size. (2h) - Intermediate
- Integrate Trivy to scan your Docker images for CVEs. (2h) - Intermediate

Week 8: Introduction to Kubernetes (K8s)

Bridge the gap between basic container orchestration and industry-standard Kubernetes.

- Install kind (Kubernetes in Docker) and spin up a local cluster. (1.5h) - Beginner
- Deploy a simple Nginx web server using a K8s Deployment and expose it via NodePort. (3h) - Intermediate

Week 9: Kubernetes Configuration & Storage

Learn how to manage state and configuration in a stateless container world.

- Deploy a stateful Redis instance using PV and PVC. (3h) - Intermediate
- Mount a ConfigMap as a volume in a Python application pod. (2h) - Intermediate

Week 10: Kubernetes Networking & Ingress

Master cluster-internal networking and exposing applications to the outside world.

- Install an Nginx Ingress Controller and configure path-based routing for two microservices. (4h) - Advanced
- Create a Network Policy to isolate a database pod from unauthorized namespaces. (3h) - Advanced

Week 11: Kubernetes Package Management with Helm

Learn Helm to template and manage complex Kubernetes applications efficiently.

- Create a custom Helm chart for your Java Spring Boot application. (4h) - Intermediate
- Deploy Prometheus and Grafana into your cluster using official Helm charts. (2h) - Intermediate

Week 12: Advanced Terraform

Upgrade Terraform skills from basic provisioning to modular, scalable infrastructure as code.

- Refactor your FoodOrdering Terraform code into reusable modules (e.g., VPC module, RDS module). (4h) - Advanced
- Configure S3 backend with DynamoDB state locking for Terraform. (2h) - Intermediate

Week 13: Configuration Management with Ansible

Introduce Ansible to manage configuration across multiple servers without agents.

- Install Ansible and write an inventory file for 3 local VMs. (1.5h) - Beginner
- Write a playbook to install Nginx, configure a custom index.html, and restart the service. (3h) - Intermediate

Week 14: Advanced Ansible & Roles

Structure Ansible code for reusability and manage sensitive data securely.

- Convert your Nginx playbook into a reusable Ansible Role. (3h) - Intermediate
- Use Ansible Vault to encrypt database passwords used in your playbooks. (2h) - Intermediate

Week 15: Advanced CI/CD with GitLab CI / GitHub Actions

Modernize CI/CD pipelines beyond Jenkins using YAML-based platforms.

- Create a GitHub Actions workflow to run Java unit tests and build a Docker image on PR. (3h) - Intermediate
- Configure OIDC between GitHub Actions and AWS to assume a role without hardcoded keys. (4h) - Advanced

Week 16: GitOps & Continuous Deployment

Implement the GitOps paradigm for Kubernetes using ArgoCD.

- Install ArgoCD on your local Kubernetes cluster. (2h) - Intermediate
- Deploy your Java application Helm chart via ArgoCD by committing changes to a Git repo. (4h) - Advanced

Week 17: DevSecOps & Pipeline Security

Shift security left by integrating automated security checks into CI/CD.

- Integrate Checkov or tfsec into your GitHub Actions pipeline to scan Terraform code. (2h) - Intermediate
- Add SonarQube quality gates to strictly block merges if coverage drops below 80%. (2h) - Intermediate

Week 18: Advanced Observability & Logging

Expand monitoring skills from basic Prometheus to centralized logging and APM.

- Deploy Fluent-bit as a DaemonSet in K8s to ship container logs to Elasticsearch. (4h) - Advanced
- Instrument your Python application with OpenTelemetry to generate distributed traces. (4h) - Advanced

Week 19: SRE Practices & Reliability

Apply Site Reliability Engineering concepts to manage system uptime and incidents.

- Define 3 SLIs and 1 SLO for your FoodOrdering platform and configure Prometheus alerts for them. (3h) - Advanced
- Write a post-mortem document for a simulated database outage. (2h) - Intermediate

Week 20: Infrastructure Cost Optimization

Learn how to monitor and optimize cloud spend, a critical skill for Senior DevOps.

- Modify your Terraform ASG to use a mixed instances policy with AWS Spot Instances. (3h) - Intermediate
- Implement AWS Budgets and configure an SNS alert when costs exceed a threshold. (2h) - Beginner

Week 21: Bridging the Gap: MLOps Basics

Address the deep learning theory gap by focusing on how to deploy and serve AI models (MLOps).

- Wrap a pre-trained HuggingFace deep learning model in a Python FastAPI application. (4h) - Intermediate
- Write a Dockerfile optimized for Python ML libraries (PyTorch/TensorFlow). (2h) - Intermediate

Week 22: Deploying AI/ML Infrastructure

Automate the deployment of the containerized ML model using DevOps tools.

- Deploy your FastAPI ML container to K8s and configure resource requests/limits appropriately. (3h) - Advanced
- Create a GitHub Actions pipeline that builds and pushes the ML image upon code changes. (3h) - Intermediate

Week 23: Chaos Engineering

Proactively test system reliability by injecting failures.

- Install Chaos Mesh in your K8s cluster and run a PodKill experiment on your ML service. (3h) - Advanced
- Observe Grafana dashboards during the chaos experiment to verify auto-recovery times. (2h) - Intermediate

Week 24: Capstone Integration & Interview Prep

Consolidate all learnings into a final project and prepare for DevOps interviews.

- Document your complete K8s GitOps architecture in a public GitHub repository. (4h) - Beginner
- Practice 5 common Senior DevOps system design interview questions. (4h) - Advanced